Unit: 6 Unit 7: Factoring
Topic: Difference of Squares

Name: ________________
Date: ___________

Quiz (Carolina Reaper)

Q1. Factor the expression: $x^2 - 16$
(A) $x^2 - 4^2$
(B) $x^2 + 4^2$
(C) $(x - 4)(x + 4)$
(D) $(x - 2)(x + 2)$
(E) $(x + 2)(x - 2)$

Q2. A shipping company has a $12 \cdot 12$ meter warehouse and wants to add a $4 \cdot 4$ meter wing. What is the total area after addition in terms of difference of squares?
(A) $(12^2 - 4^2)$
(B) $12^2 + 4^2$
(C) $(12^2 + 4^2)$
(D) $(12 + 4)(12 - 4)$
(E) $(12 - 4)(12 + 4)$

Q3. What is the factored form of $49x^2 - 9$?
(A) $(7x - 3)(7x + 3)$
(B) $(9x - 7)(9x + 7)$
(C) $(7x - 9)(7x + 9)$
(D) $(7x + 3)(7x - 3)$
(E) $(3x - 7)(3x + 7)$

Q4. A traffic light at an intersection changes every 9 seconds. A second traffic light changes every 4 seconds. How many seconds will pass before both traffic lights change together again in terms of difference of squares?
(A) $9^2 - 4^2$
(B) (9 + 4)(9 - 4)
(C) $9^2 + 4^2$
(D) $36 \cdot 4$
(E) $9 \cdot 4$

Q5. A delivery truck travels at 25 km/h. A second truck travels at 16 km/h. What is the difference in their distances traveled after 2 hours in terms of difference of squares?
(A) $(25^2 - 16^2) \cdot 2$
(B) $(25^2 + 16^2)$
(C) $(25 + 16)(25 - 16) \cdot 2$
(D) $(25 - 16)(25 + 16) \cdot 2$
(E) $(25 + 16) \cdot 2$

Q6. A cargo plane flies 121 km to a destination and then returns. What is the total distance traveled by the plane in terms of difference of squares?
(A) $(11^2 - 0^2) \cdot 2$
(B) $(11^2 + 0^2) \cdot 2$
(C) $(11^2 - 11^2)$
(D) $(11^2 + 11^2)$
(E) $11^2 \cdot 2$

Q7. A ship travels from point A to point B in 4 hours and returns in 9 hours. What is the difference in their travel times in terms of difference of squares?
(A) $9^2 - 4^2$
(B) $9^2 + 4^2$
(C) $(9 - 4)(9 + 4)$
(D) $(9 + 4)(9 - 4)$
(E) $9 + 4$

Q8. A logistics company has 25 trucks and 9 cars. What is the difference in the squares of their quantities?
(A) $25^2 - 9^2$
(B) $(25 + 9)(25 - 9)$
(C) $25^2 + 9^2$
(D) $25 + 9$
(E) $25 - 9$

Open Ended Questions (FRQ)

Q9. A transportation company wants to add a rectangular wing to their existing warehouse. If the length and width of the existing warehouse are 15 meters and 8 meters respectively, and the length and width of the wing are 12 meters and 5 meters respectively, find the total area after addition using the difference of squares formula. Show all work.

Q10. Two delivery trucks start from the same point and travel to the same destination. One truck travels 11 km/h faster than the other. If the slower truck takes 10 hours to reach the destination, how many hours does the faster truck take to reach the destination? Use difference of squares to find the time taken by the faster truck. Show all work.

Chef's Tips

- Read each question carefully before answering.
- Use the difference of squares formula whenever applicable.
- Show all work in the free response questions.